

Introduction

Aircraft are now more widely used than ever before with no sign of stopping. As such, researchers all constantly looking for methods of improving their efficiency. Improving the efficiency of aircraft reduces the fuel consumption and improves overall performance. One method of improving aircraft efficiency is to reduce their air resistance.

Background

- Denticles are small enamel structures that help sharks glide through the water.
- Denticles work in a similar way to the dimples on golf balls.
- Since both water and air are fluids, scientists wondered if the same denticles that help sharks move through water, could be used to help aircraft move through air.
- Synthetically recreated denticles have been applied to cargo planes and showed to reduce air resistance.
- Synthetic denticles are expensive and laborious to apply and maintain, resulting in them not being widely used.

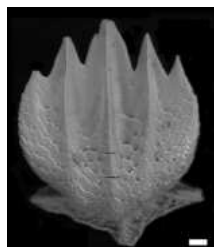


Figure 1
Single denticle



Figure 2
Shark skin under microscope

Purpose & Hypothesis

My project's goal was to optimize the shape of synthetically replicated denticles, aiming to make them worth the cost and labor of being implemented. I hypothesized that denticles with a large L1 value would result in the lowest wind resistance, similar to how the elongated shape of professional bike helmets allows riders to reduce their drag.

Optimizing the Structure of Synthetically Replicated Shark Denticles to Reduce Drag on High Speed Aircraft

Method

In order to optimize the geometry of the synthetically replicated denticles for reducing the air resistance of aircraft, I observed the denticle of a Mako shark and split it into 5 distinct variables as seen in figure 3.

I chose to manipulate the L1 variable, as it would likely have the greatest effect on aerodynamic performance due to the fact that denticles located on different parts of a shark's body exhibited the most change in the L1 region, meaning the geometry of the L1 variable likely has the greatest effect on drag force. I then decided how I would manipulate the L1 variable, and decided on a specific series of modifications.

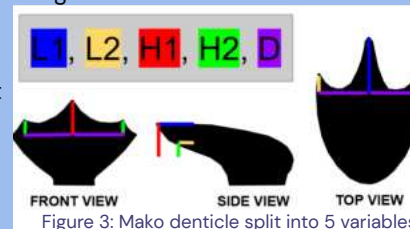
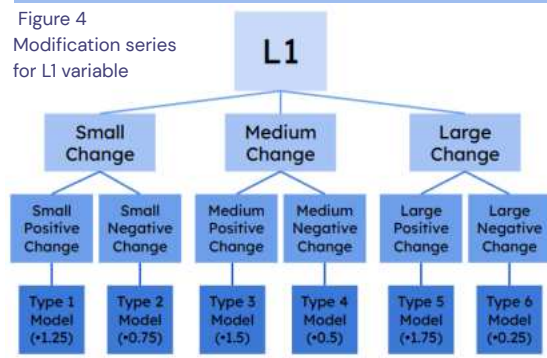


Figure 3: Mako denticle split into 5 variables

Figure 4
Modification series for L1 variable



To test the denticles, they were affixed in batches of 37 to a section of an airfoil. The denticle-laden airfoils were referred to as a model. All 37 denticles in a single model were the same, but each of the models's denticles varied in their L1 geometry. The airfoil and denticles were modeled in blender, with the original denticle being based off of a Mako's due to it being the fastest shark. The models were then printed using a resin printer and tested in a wind tunnel with a force sensor. A type O model with an unchanged L1 value and a type N model with no denticles were also tested and served as controls.

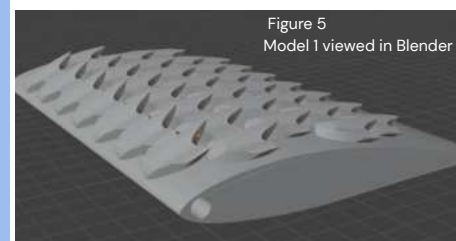


Figure 5
Model 1 viewed in Blender

Data

After testing, the following data was collected and a graph was made as seen in figure 6 and 7.

Model Type	Type N	Type 0	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6
Wind Speed (ft/min * 1000)	0	0	0	0	0	0	0	0
1	2	2	0	1	1	1	2	1
1.5	3	4	1	4	1	2	3	2
2	4	8	2	8	2	3	4	3
3	6	9	6	18	5	6	8	6
4	12	12	10	22	11	11	13	8

Figure 6: Drag force data for all models at different wind tunnel speeds

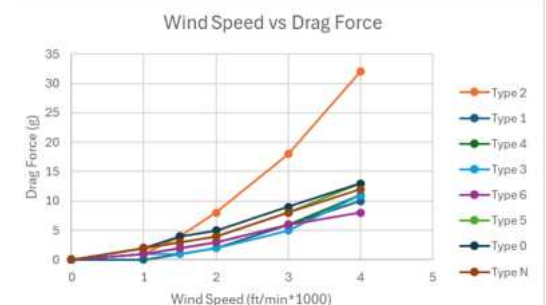


Figure 7: Graphed data from figure 6

Results

The data collected in my experiment was unable to verify nor deny my hypothesis, resulting in my conclusion being that changes in the L1 value resulted in no significant changes to wind resistance. I believe that I would have seen greater change in wind resistance among groups if the denticles on the models were scaled down, but I was unable to do so due to the constraints of the resin printer. Additionally, choosing to alter different variables would likely offer different results, but I was unable to do so with the provided time.

Sources

- Figure 1: <https://saveourseas.com/update/a-sharks-scale/>
Figure 2: <https://ocean.si.edu/ocean-life/sharks-rays/biomimicry-shark-denticles>
Figure 3: Image created by finalist using google slides
Figure 4: Image created by finalist using google slides
Figure 5: Screenshot taken by finalist in Blender
Figure 6: Screenshot taken by finalist in excel
Figure 7: Screenshot taken by finalist in excel